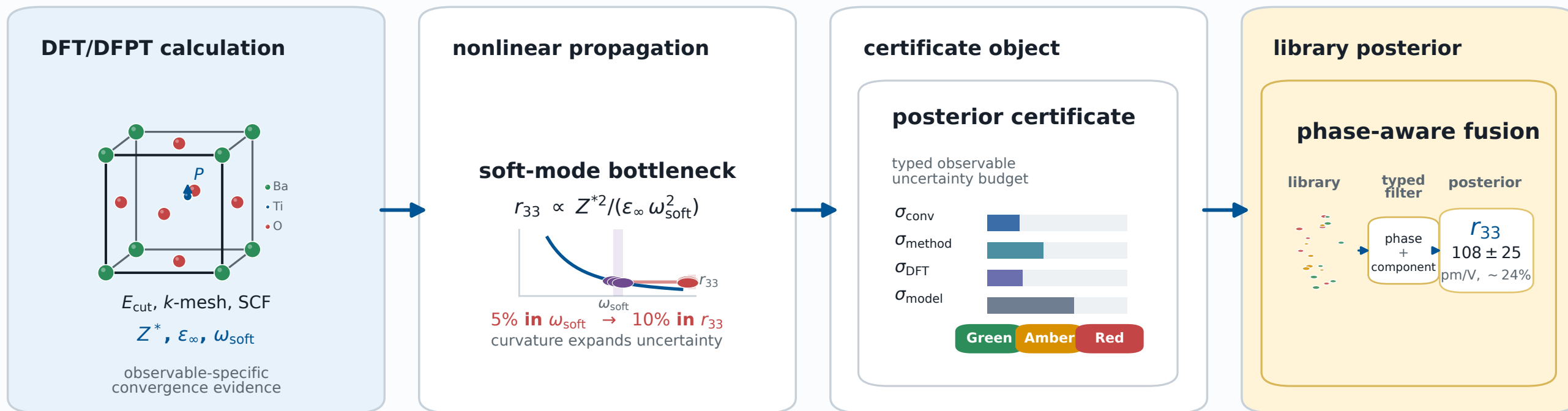




What travels with each value

source labels

observable type

quality tier

propagation rule

machine-readable record

```
{ "property": "r33_iso",
  "value": "108 pm/V",
  "sigma_rel": "28.4%",
  "components": [
    "conv", "method",
    "DFT", "model"
  ],
  "quality_tier": "Red"
}
```

Outcome for reusable materials data

single calculation: 28.4% Red-tier response

phase-filtered fusion: 108 ± 25 pm/V

floor-limited uncertainty: ~24% toward ~22%

Core idea: DFT values should travel with calibrated uncertainty, provenance, and validity scope.